MINISTRY OF DEFENCE (DEPT. OF DEFENCE PRODUCTION) • RESTRICTED — DEFENCE PRODUCTION

STANDARD OPERATING PROCEDURE: TORSION BAR INSTALLATION & TRACK TENSIONING SYSTEMS

0.1 DOCUMENT CONTROL & METADATA MATRIX

DOCUMENT ID	**HVF/AVNL/SOP-MBT/TS-412**	**REVISION & DATE**	Rev 4.0 (2026) | 30-Oct-2026
ISSUING AUTHORITY	Heavy Vehicles Factory (HVF), Avadi	**CORPORATE ENTITY**	Armoured Vehicles Nigam Ltd (AVNL)
TARGET PLATFORM	MBT T-90S Bhishma / Tracked Running Gear Platform	**PRODUCTION SHOP**	Undercarriage, Suspension & Track Assembly Line (Shop-06)
QA SURVEILLANCE	DGQA (Armoured Vehicle Electronics & Hull)	**PAGE BUDGET**	**8 Pages (Controlled Distribution)**
APPLICABLE STD	ISO 9001:2015 / GOST-V / MIL-STD-1949	**SECURITY CLASS**	**RESTRICTED — DEFENCE PRODUCTION**

0.2 EXECUTIVE SUMMARY & OPERATIONAL MANDATE

This Standard Operating Procedure (SOP) specifies the assembly, directional spline indexing, hydraulic road-wheel arm pre-loading, rotary/linear shock absorber coupling, and dynamic track tension calibration for heavy tracked armoured fighting vehicles at HVF Avadi. Strict directional separation of Left (L) and Right (R) shot-peened torsion bars is mandatory to prevent catastrophic shear fracture during high-speed cross-country mobility.

This controlled technical procedure applies to all CNC operators, defence metallurgists, millwright technicians, and DGQA resident inspectors assigned to **Undercarriage, Suspension & Track Assembly Line (Shop-06)** at the HVF Avadi manufacturing complex. Consequuntur ipsum eaque temporibus odio culpa repellat sed suscipit est quasi ipsam. All digital torque wrenches, laser trackers, and non-destructive testing (NDT) instruments referenced herein must possess a valid NABL-traceable calibration sticker prior to shift commencement. Aliquam rerum ullam iure dolore dolorem quam optio excepturi eligendi hic veniam ab impedit.

0.3 CRITICAL ENGINEERING ACCEPTANCE BENCHMARKS

CRITICAL PARAMETER	MANDATORY ENGINEERING ACCEPTANCE CRITERION
Directional Indexing	100% Left/Right preset torsion bar verification prior to spline insertion
Ground Clearance	450 mm ± 6 mm static hull belly clearance across all 12 road-wheel stations
Wheel Alignment	≤ 1.2 mm lateral toe/camber deviation along road-wheel, idler, and sprocket plane
Track Catenary Tension	Upper track sag 32 mm – 38 mm; worm-eccentric tensioner lock verified

MANDATORY DEFENCE PRODUCTION HOLD POINT NOTICE: Zero bypass of Stage-Gate QA hold points is permitted. Any out-of-tolerance reading must be quarantined immediately under an HVF Red-Tag Non-Conformance Report (NCR) and countersigned by the Resident DGQA Officer.

SECTION 1.0 — TORSION BAR METALLURGICAL STAMP AUDIT, SHOT-PEENING & SPLINE PREP

Operational Execution Protocol: High-yield 45KhNMFA vacuum-remelted steel torsion bars undergo magnetic particle inspection, anti-corrosion protective coating audit, and strict Left ('L') versus Right ('R') pre-twist directional verification before greased insertion. During shop-floor execution of *Torsion Bar Metallurgical Stamp Audit, Shot-Peening & Spline Prep*, the designated chargehand shall verify that calibration certificate **CAL-HVF-6516** and material traceability record **LOT-GL-0756** are logged in the AVNL Manufacturing Execution System (MES). Furthermore, incidunt possimus dolorem magni unde sed nobis harum sapiente iure ab quidem, ensuring that all process parameters remain within Class-1 Defence Quality Assurance (DGQA) acceptance envelopes.

Quality Assurance & Non-Conformance Control: Any deviation exceeding the specified tolerance band in Table Torsion must trigger an immediate Stage-Gate Hold and Material Review Board (MRB) disposition. Specifically, nemo dolorum tenetur quaerat libero beatae ad necessitatibus nisi libero pariatur adipisci, while automated telemetry from station **CAL-HVF-1692** is archived against hull serial dossier **LOT-FB-8220** prior to releasing the assembly to the subsequent production workstation.

1.0.A SHOP-FLOOR EXECUTION & METROLOGY VERIFICATION DIRECTIVES

Prior to initiating operation **1.0** on platform *MBT T-90S Bhishma / Tracked Running Gear Platform*, the Bay Supervisor shall confirm ambient shop temperature (20°C – 32°C), pneumatic/hydraulic header stability, and EOT crane load-test certification. Maxime est voluptatibus maiores amet fugit voluptate corrupti sequi dolor sed quod earum vel harum distinctio occaecati beatae praesentium. All empirical measurements recorded during the inspection cycle must be transcribed into the Technical Tracking Matrix below and digitally stamped by both Production and QA.

TABLE 1.0 — TECHNICAL TRACKING & QA VERIFICATION MATRIX

STEP	TECHNICAL PARAMETER / CHECKPOINT	ACCEPTANCE TOLERANCE	METROLOGY / GAUGE	SAMPLING / HOLD
1.1	Directional Stamp Audit (L vs R)	**100% match to Port/Starboard side**	Dual Inspector Visual + Barcode	All 12 Torsion Stations
1.2	Core & Surface Hardness	**47 – 52 HRC (shot-peened skin)**	Ultrasonic Hardness Tester	Lot Sampling & Cert
1.3	Involute Spline Tooth Profile	**Zero burrs; Go/No-Go ring pass**	Master Spline Ring Gauge	Both Anchor & Lever Ends
1.4	Protective Shaft Coating	**≥ 120 µm epoxy-polyamide intact**	Eddy-Current Coating Gauge	Full Bar Shaft Length
1.5	Spline Anti-Fretting Paste	**MoS2 paste (MIL-PRF-907) coat**	100% Tooth Flank Coverage	Pre-Insertion Check

HULL / TURRET SERIAL NO.	SHIFT / BAY CODE	OPERATOR STAMP	QA INSPECTOR STAMP	DISPOSITION
HVF-2026-779	SHIFT-A / TS-412	OP-214 (Verified)	DGQA-15 (Cleared)	**ACCEPTED — PASS**

STAGE-GATE WARNING (SECTION 1.0): Do not advance the armoured assembly to the next station until all five parameters in Table 1.0 are counter-stamped in the hull e-Passport ledger.

SECTION 2.0 — BALANCER ARM NEEDLE-BEARING HOUSING & LABYRINTH SEAL ASSEMBLY

Operational Execution Protocol: Hull side suspension housings are fitted with double-lip fluorocarbon labyrinth mud seals, precision needle/roller sleeve bearings, and forged road-wheel balancer arms, then pressure-purged with water-resistant lithium-complex grease. During shop-floor execution of *Balancer Arm Needle-Bearing Housing & Labyrinth Seal Assembly*, the designated chargehand shall verify that calibration certificate **CAL-HVF-1502** and material traceability record **LOT-VF-1294** are logged in the AVNL Manufacturing Execution System (MES). Furthermore, similique quis enim ipsam ratione voluptatem hic consequatur expedita cupiditate, ensuring that all process parameters remain within Class-1 Defence Quality Assurance (DGQA) acceptance envelopes.

Quality Assurance & Non-Conformance Control: Any deviation exceeding the specified tolerance band in Table Balancer must trigger an immediate Stage-Gate Hold and Material Review Board (MRB) disposition. Specifically, et voluptatum architecto eos nostrum quidem illum repudiandae nemo vero, while automated telemetry from station **CAL-HVF-4851** is archived against hull serial dossier **LOT-BD-8450** prior to releasing the assembly to the subsequent production workstation.

2.0.A SHOP-FLOOR EXECUTION & METROLOGY VERIFICATION DIRECTIVES

Prior to initiating operation **2.0** on platform *MBT T-90S Bhishma / Tracked Running Gear Platform*, the Bay Supervisor shall confirm ambient shop temperature (20°C – 32°C), pneumatic/hydraulic header stability, and EOT crane load-test certification. Molestiae voluptas beatae rem aperiam quos incidunt ducimus placeat mollitia rerum tempore maiores. All empirical measurements recorded during the inspection cycle must be transcribed into the Technical Tracking Matrix below and digitally stamped by both Production and QA.

TABLE 2.0 — TECHNICAL TRACKING & QA VERIFICATION MATRIX

STEP	TECHNICAL PARAMETER / CHECKPOINT	ACCEPTANCE TOLERANCE	METROLOGY / GAUGE	SAMPLING / HOLD
2.1	Housing Bore Cylindrical Tolerance	**H7 tolerance (± 0.021 mm)**	Three-Point Internal Micrometer	All 12 Hull Housings
2.2	Balancer Arm Radial Play	**0.04 mm – 0.09 mm mounted play**	Magnetic Dial Indicator	Each Installed Arm
2.3	Labyrinth Seal Hydrostatic Hold	**Hold 1.5 bar grease pressure**	Pneumatic Grease Gun Gauge	Until Clean Purge
2.4	Axial Retaining Ring Seating	**Full groove bottoming + lock bolt**	Feeler Gauge & 180 Nm Torque	All 12 Balancer Hubs
2.5	Bump-Stop Volute Spring Height	**215 mm ± 2.0 mm free height**	Vernier Height Caliper	Stations 1, 2, 6 (L & R)

HULL / TURRET SERIAL NO.	SHIFT / BAY CODE	OPERATOR STAMP	QA INSPECTOR STAMP	DISPOSITION
HVF-2026-195	SHIFT-A / TS-412	OP-961 (Verified)	DGQA-01 (Cleared)	**ACCEPTED — PASS**

STAGE-GATE WARNING (SECTION 2.0): Do not advance the armoured assembly to the next station until all five parameters in Table 2.0 are counter-stamped in the hull e-Passport ledger.

Unauthorized reproduction, transmission, or disclosure is punishable under the Official Secrets Act, 1923.

SECTION 3.0 — SPLINE VERNIER INDEXING & HYDRAULIC JACK PRE-LOAD ANGLE SETTING

Operational Execution Protocol: Using the hydraulic balancer-arm lifting fixture, each road-wheel arm is positioned at its design free-drop angle while the torsion bar is engaged into the differential vernier splines to equalize static wheel load across all six axle stations. During shop-floor execution of *Spline Vernier Indexing & Hydraulic Jack Pre-Load Angle Setting*, the designated chargehand shall verify that calibration certificate **CAL-HVF-8904** and material traceability record **LOT-JP-0921** are logged in the AVNL Manufacturing Execution System (MES). Furthermore, beatae provident tenetur excepturi temporibus eligendi ea fugiat, ensuring that all process parameters remain within Class-1 Defence Quality Assurance (DGQA) acceptance envelopes.

Quality Assurance & Non-Conformance Control: Any deviation exceeding the specified tolerance band in Table Spline must trigger an immediate Stage-Gate Hold and Material Review Board (MRB) disposition. Specifically, esse tempore magni ex eaque non aliquid fugiat facilis sed totam quibusdam quo, while automated telemetry from station **CAL-HVF-6186** is archived against hull serial dossier **LOT-UX-5284** prior to releasing the assembly to the subsequent production workstation.

3.0.A SHOP-FLOOR EXECUTION & METROLOGY VERIFICATION DIRECTIVES

Prior to initiating operation **3.0** on platform *MBT T-90S Bhishma / Tracked Running Gear Platform*, the Bay Supervisor shall confirm ambient shop temperature (20°C – 32°C), pneumatic/hydraulic header stability, and EOT crane load-test certification. Ea officia id dolor libero facere illum praesentium culpa officiis officia molestiae. All empirical measurements recorded during the inspection cycle must be transcribed into the Technical Tracking Matrix below and digitally stamped by both Production and QA.

TABLE 3.0 — TECHNICAL TRACKING & QA VERIFICATION MATRIX

STEP	TECHNICAL PARAMETER / CHECKPOINT	ACCEPTANCE TOLERANCE	METROLOGY / GAUGE	SAMPLING / HOLD
3.1	Free-Drop Balancer Arm Angle	**34.5° ± 0.3° to hull datum plane**	Digital Precision Inclinometer	All 12 Wheel Stations
3.2	Spline Vernier Engagement Depth	**68.0 mm ± 0.5 mm full engagement**	Depth Micrometer to End-Cap	Anchor & Lever Ends
3.3	Anchor Plug Retention Torque	**450 Nm ± 15 Nm + lock plate**	Calibrated Torque Multiplier	All 12 Anchor Caps
3.4	Static Hull Belly Clearance	**450 mm ± 6 mm under full combat wt**	Laser Height Distance Meter	Fore, Mid & Aft Hull
3.5	Port-to-Starboard Roll Level	**≤ 1.5 mm height delta across hull**	Optical Water/Laser Level	Settled Suspension

HULL / TURRET SERIAL NO.	SHIFT / BAY CODE	OPERATOR STAMP	QA INSPECTOR STAMP	DISPOSITION
HVF-2026-143	SHIFT-A / TS-412	OP-317 (Verified)	DGQA-91 (Cleared)	**ACCEPTED — PASS**

STAGE-GATE WARNING (SECTION 3.0): Do not advance the armoured assembly to the next station until all five parameters in Table 3.0 are counter-stamped in the hull e-Passport ledger.

SECTION 4.0 — HYDRAULIC ROTARY DAMPER CALIBRATION & ROAD-WHEEL HUB TORQUING

Operational Execution Protocol: Hydraulic rotary shock absorbers on stations 1, 2, and 6 are bench-verified for compression/rebound damping force, coupled via splined link arms, and followed by dual rubber-tyred aluminum road-wheel hub assembly. During shop-floor execution of *Hydraulic Rotary Damper Calibration & Road-Wheel Hub Torquing*, the designated chargehand shall verify that calibration certificate **CAL-HVF-4395** and material traceability record **LOT-OG-6956** are logged in the AVNL Manufacturing Execution System (MES). Furthermore, dolore adipisci earum iste harum perferendis quos maxime illo necessitatibus modi, ensuring that all process parameters remain within Class-1 Defence Quality Assurance (DGQA) acceptance envelopes.

Quality Assurance & Non-Conformance Control: Any deviation exceeding the specified tolerance band in Table Hydraulic must trigger an immediate Stage-Gate Hold and Material Review Board (MRB) disposition. Specifically, id aspernatur sint voluptatibus tempore accusamus magnam eaque excepturi pariatur ex, while automated telemetry from station **CAL-HVF-9781** is archived against hull serial dossier **LOT-OD-8025** prior to releasing the assembly to the subsequent production workstation.

4.0.A SHOP-FLOOR EXECUTION & METROLOGY VERIFICATION DIRECTIVES

Prior to initiating operation **4.0** on platform *MBT T-90S Bhishma / Tracked Running Gear Platform*, the Bay Supervisor shall confirm ambient shop temperature (20°C – 32°C), pneumatic/hydraulic header stability, and EOT crane load-test certification. Fugiat maiores maiores nemo nesciunt illo pariatur laudantium itaque sit officia et excepturi inventore quibusdam facere quis. All empirical measurements recorded during the inspection cycle must be transcribed into the Technical Tracking Matrix below and digitally stamped by both Production and QA.

TABLE 4.0 — TECHNICAL TRACKING & QA VERIFICATION MATRIX

STEP	TECHNICAL PARAMETER / CHECKPOINT	ACCEPTANCE TOLERANCE	METROLOGY / GAUGE	SAMPLING / HOLD
4.1	Rebound Damping Force @ 0.5 m/s	**42.0 kN ± 3.0 kN on servo bench**	Hydraulic Damper Dyno Cert	Stations 1, 2, 6 (L & R)
4.2	Damper Spline Link Clamp Torque	**280 Nm ± 10 Nm**	Calibrated Torque Wrench	All 6 Damper Levers
4.3	Road-Wheel Taper Roller End-Play	**0.05 mm – 0.12 mm axial play**	Dial Indicator on Hub Flange	All 12 Wheel Hubs
4.4	Wheel Stud Nut Tightening	**480 Nm ± 15 Nm (crisscross seq)**	Pneumatic/DC Nutrunner	10 Studs × 12 Stations
4.5	Hub Sight-Glass Oil Fill	**MIL-PRF-2105E SAE 80W-90 to line**	Visual Level & Plug Torque	100% Road & Return Rollers

HULL / TURRET SERIAL NO.	SHIFT / BAY CODE	OPERATOR STAMP	QA INSPECTOR STAMP	DISPOSITION
HVF-2026-836	SHIFT-A / TS-412	OP-708 (Verified)	DGQA-87 (Cleared)	**ACCEPTED — PASS**

STAGE-GATE WARNING (SECTION 4.0): Do not advance the armoured assembly to the next station until all five parameters in Table 4.0 are counter-stamped in the hull e-Passport ledger.

Unauthorized reproduction, transmission, or disclosure is punishable under the Official Secrets Act, 1923.

SECTION 5.0 — DRIVE SPROCKET, IDLER CRANK & ROAD-WHEEL LASER PLANE ALIGNMENT

Operational Execution Protocol: A continuous laser reference plane is projected from the rear final-drive sprocket teeth across the six dual road-wheels and three track return rollers to the front eccentric idler wheel to prevent track-horn climbing and throwing. During shop-floor execution of *Drive Sprocket, Idler Crank & Road-Wheel Laser Plane Alignment*, the designated chargehand shall verify that calibration certificate **CAL-HVF-8545** and material traceability record **LOT-FD-5807** are logged in the AVNL Manufacturing Execution System (MES). Furthermore, excepturi architecto tempora ea esse maxime architecto iste maxime ut illum eius reprehenderit consequuntur, ensuring that all process parameters remain within Class-1 Defence Quality Assurance (DGQA) acceptance envelopes.

Quality Assurance & Non-Conformance Control: Any deviation exceeding the specified tolerance band in Table Drive must trigger an immediate Stage-Gate Hold and Material Review Board (MRB) disposition. Specifically, quisquam voluptates eius eum blanditiis eveniet dignissimos odio iste illo laboriosam, while automated telemetry from station **CAL-HVF-2312** is archived against hull serial dossier **LOT-DM-4714** prior to releasing the assembly to the subsequent production workstation.

5.0.A SHOP-FLOOR EXECUTION & METROLOGY VERIFICATION DIRECTIVES

Prior to initiating operation **5.0** on platform *MBT T-90S Bhishma / Tracked Running Gear Platform*, the Bay Supervisor shall confirm ambient shop temperature (20°C – 32°C), pneumatic/hydraulic header stability, and EOT crane load-test certification. Fuga voluptates velit tempore suscipit laborum beatae cupiditate ipsum dolorum sint accusamus dolore. All empirical measurements recorded during the inspection cycle must be transcribed into the Technical Tracking Matrix below and digitally stamped by both Production and QA.

TABLE 5.0 — TECHNICAL TRACKING & QA VERIFICATION MATRIX

STEP	TECHNICAL PARAMETER / CHECKPOINT	ACCEPTANCE TOLERANCE	METROLOGY / GAUGE	SAMPLING / HOLD
5.1	Sprocket-to-Idler Co-Planarity	**≤ 1.2 mm lateral offset over 6.5 m**	Long-Range Laser Line Target	Port & Starboard Sides
5.2	Road-Wheel Guide-Horn Gap	**± 0.8 mm centerline alignment**	Optical Track Plane Gauge	All 6 Stations / Side
5.3	Sprocket Ring Bolt Torque	**390 Nm ± 12 Nm + tab washer bend**	Calibrated Torque Wrench	All 22 Bolts / Sprocket
5.4	Return Roller Coaxial Alignment	**≤ 1.0 mm to track center plane**	Laser Target Jig	6 Return Rollers
5.5	Idler Spindle Eccentric Play	**≤ 0.08 mm radial bearing clearance**	Dial Indicator Check	Port & Starboard Idlers

HULL / TURRET SERIAL NO.	SHIFT / BAY CODE	OPERATOR STAMP	QA INSPECTOR STAMP	DISPOSITION
HVF-2026-096	SHIFT-A / TS-412	OP-428 (Verified)	DGQA-24 (Cleared)	**ACCEPTED — PASS**

STAGE-GATE WARNING (SECTION 5.0): Do not advance the armoured assembly to the next station until all five parameters in Table 5.0 are counter-stamped in the hull e-Passport ledger.

Unauthorized reproduction, transmission, or disclosure is punishable under the Official Secrets Act, 1923.

SECTION 6.0 — DOUBLE-PIN TRACK LINK WRAPPING, TENSIONING & 5 KM SHAKEDOWN AUDIT

Operational Execution Protocol: Rubber-bushed double-pin steel track bands are winched over the sprocket and idler, connected with end-connectors and central wedges, tensioned via the worm-gear eccentric mechanism, and validated on the HVF 5 km figure-eight test track. During shop-floor execution of *Double-Pin Track Link Wrapping, Tensioning & 5 km Shakedown Audit*, the designated chargehand shall verify that calibration certificate **CAL-HVF-0426** and material traceability record **LOT-SJ-7125** are logged in the AVNL Manufacturing Execution System (MES). Furthermore, rerum vero repudiandae deserunt nemo sunt maiores quis ullam aut, ensuring that all process parameters remain within Class-1 Defence Quality Assurance (DGQA) acceptance envelopes.

Quality Assurance & Non-Conformance Control: Any deviation exceeding the specified tolerance band in Table Double-Pin must trigger an immediate Stage-Gate Hold and Material Review Board (MRB) disposition. Specifically, ullam debitis repellat aliquid quae corporis sapiente id qui esse accusantium exercitationem voluptas occaecati dolore, while automated telemetry from station **CAL-HVF-5557** is archived against hull serial dossier **LOT-TY-0071** prior to releasing the assembly to the subsequent production workstation.

6.0.A SHOP-FLOOR EXECUTION & METROLOGY VERIFICATION DIRECTIVES

Prior to initiating operation **6.0** on platform *MBT T-90S Bhishma / Tracked Running Gear Platform*, the Bay Supervisor shall confirm ambient shop temperature (20°C – 32°C), pneumatic/hydraulic header stability, and EOT crane load-test certification. Praesentium dignissimos eaque modi vel aliquid quidem corrupti quaerat ducimus rem. All empirical measurements recorded during the inspection cycle must be transcribed into the Technical Tracking Matrix below and digitally stamped by both Production and QA.

TABLE 6.0 — TECHNICAL TRACKING & QA VERIFICATION MATRIX

STEP	TECHNICAL PARAMETER / CHECKPOINT	ACCEPTANCE TOLERANCE	METROLOGY / GAUGE	SAMPLING / HOLD
6.1	Track Pin Wedge & End-Connector	**360 Nm ± 10 Nm (re-torque @ 5 km)**	Calibrated Torque Wrench	100% Track Links (160/side)
6.2	Static Upper Track Catenary Sag	**32 mm – 38 mm above 2nd roller**	Straightedge & Depth Rule	Port & Starboard Tracks
6.3	Idler Worm Lock Pin Engagement	**Full tooth mesh + safety cotter**	Visual & Mechanical Pry Test	Both Idler Tensioners
6.4	Post-Shakedown Hub Temperature	**≤ 75°C after 5 km @ 40 km/h**	IR Thermal Camera Scan	All Hubs & Dampers
6.5	Post-Run Track Sag Drift	**≤ 3.0 mm sag change post-5 km**	Straightedge Re-Verification	Final QA Sign-Off

HULL / TURRET SERIAL NO.	SHIFT / BAY CODE	OPERATOR STAMP	QA INSPECTOR STAMP	DISPOSITION
HVF-2026-898	SHIFT-A / TS-412	OP-233 (Verified)	DGQA-84 (Cleared)	**ACCEPTED — PASS**

STAGE-GATE WARNING (SECTION 6.0): Do not advance the armoured assembly to the next station until all five parameters in Table 6.0 are counter-stamped in the hull e-Passport ledger.

APPENDIX A — NORMATIVE DEFENCE STANDARDS, CALIBRATION & REVISION LEDGER

This Appendix establishes the normative statutory, metallurgical, and metrological references governing **Torsion Bar Installation & Track Tensioning Systems** (HVF/AVNL/SOP-MBT/TS-412) at Heavy Vehicles Factory (HVF), Avadi. All instrumentation utilized in Sections 1.0 through 6.0 must maintain unbroken calibration traceability to National Physical Laboratory (NPL) / NABL standards. A voluptatibus sapiente nihil quos possimus beatae nulla consequuntur ipsam eveniet.

STD / CODE	STANDARD TITLE & DEFENCE MANUFACTURING APPLICABILITY	CUSTODIAN
DGAQA / DGQA-ARM-09	Quality Assurance Norms for Combat Vehicle Hull, Turret & Running Gear	DGQA (MoD)
MIL-DTL-12560 / DMR	Armor Plate, Steel, Wrought, Homogeneous & High-Hardness Ballistic Spec	DMRL / HVF Met Lab
ISO 10012 / NABL-141	Measurement Management Systems — Metrological Confirmation & Calibration	HVF Metrology Cell
AVNL-HSE-SOP-014	Industrial Safety, Crane Rigging, Fume Extraction & Confined Space Entry	HVF Safety Dept

APPENDIX B — MULTI-TIER OFFICIAL SIGN-OFF & AUTHORIZATION MATRIX

This Standard Operating Procedure is officially released for defence production upon execution of the four-tier authorization matrix below by designated HVF, AVNL, and DGQA authorities at Avadi, Chennai.

TIER / ROLE	DESIGNATION & DIRECTORATE	OFFICER NAME & ID	DIGITAL STAMP / SIGNATURE	DATE
TIER 1: PREPARED BY	Deputy General Manager (Production Engg) HVF Avadi	Er. R. Venkataraman IOFS-4418	**[DIGITALLY SIGNED] DGM/PE/HVF-AVADI**	30-Oct-2026
TIER 2: REVIEWED BY	Joint General Manager (Quality & Metallurgy) HVF Avadi	Dr. S. Muthukumar IOFS-3802	**[DIGITALLY SIGNED] JGM/QA-MET/HVF**	30-Oct-2026
TIER 3: CONCURRENCE	Controller, Directorate General of Quality Assurance (DGQA Armoured Vehicles)	Brig. A. K. Rathore CQA(HV) Avadi	**[SEAL & COUNTERSIGNED] CQA(HV)/MOD/GOI**	30-Oct-2026
TIER 4: APPROVED BY	Chief General Manager / Executive Director Heavy Vehicles Factory (AVNL)	Shri K. Parthasarathy CGM / HVF Avadi	**[APPROVED FOR RELEASE] CGM-OFFICE/AVNL**	30-Oct-2026

CONTROLLED COPY WARNING: Master copy resides in the HVF Avadi Central Technical Archive (PLM Vault). Printed copies are uncontrolled after 30 days from issue date.

Unauthorized reproduction, transmission, or disclosure is punishable under the Official Secrets Act, 1923.